
Code-switching in Mandarin and English Definite Noun Phrases

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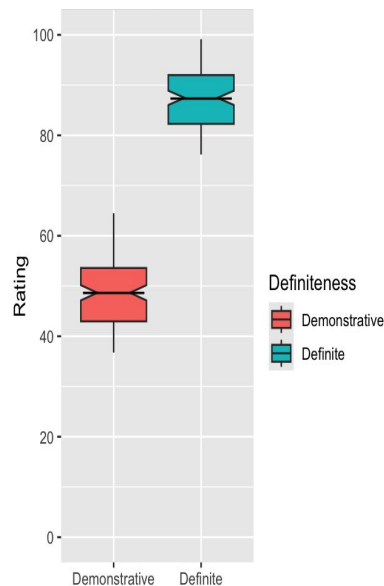


Determiners in Anaphoric (across discourse) DPs

- ❖ Definite determiner
 - I met a boy. The boy was funny.
- ❖ Demonstrative determiner
 - I met a boy. That boy was funny.

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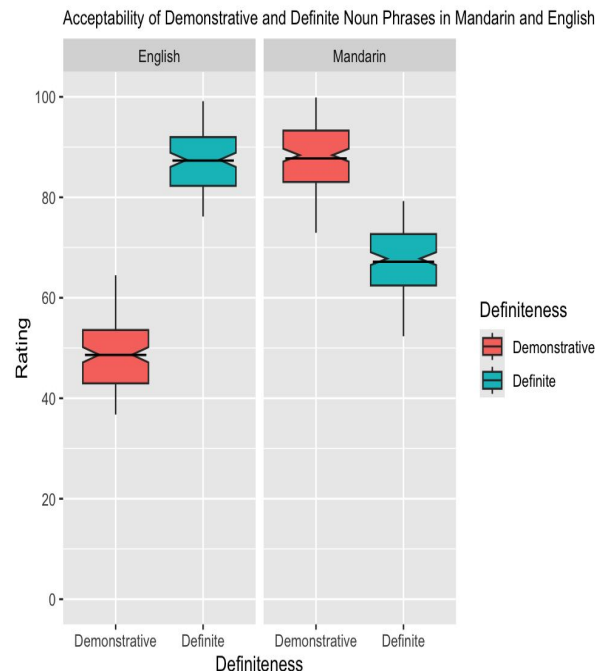


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- ❖ Definite determiner
 - I met a boy. The boy was funny.
- ❖ Demonstrative determiner
 - I met a boy. That boy was funny.
- ❖ Null determiner
 - I met a boy. ∅ boy was funny. (Not good in English, but good in Mandarin)
 - yi-ge nanhai zoujin-le jiaoshi.
a-CLF boy walk-in-PST classroom.
'A boy entered the classroom'
 - ∅ nanhai zuozai-le qian pai
boy sit-PST front row
'The boy sat down in the front row.'

Anaphoric determiners in Mandarin vs. English

- ❖ Saha et al., 2024: English prefers definite determiner over demonstrative determiner in this context, while Mandarin prefers the “demonstrative” over the bare noun:
 - English: A boy entered the classroom. That/the boy sat down in the front row.
 - Mandarin: yi-ge nanhai zoujin-le jiaoshi.
a-CLF boy walk-in-PST classroom.
'A boy entered the classroom'
nage/∅ nanhai zuozai-le qian pai
that boy sit-PST front row
'The boy sat down in the front row.'





Research Question

- ❖ Prior research on code-switching between the determiner and the noun within noun phrases has often focused on cases of gender features, for example, cases where one language has lexical gender and another does not (e.g. Spanish/English: *el nino* /*the boy*/*el boy*/*the nino*)
- ❖ Some conclude from this that code-switching is best in these cases when the determiner matches the matrix language (Myers-Scotton, 1993); others have concluded that code-switching is best when the determiner is from the language with the largest number of uninterpretable features (e.g., MacSwan 2005).
 - La cow > The vaca (Spanish-English code-switching)
- ❖ This study brings in new perspectives by focusing on the use of demonstrative determiners in Mandarin-English code-switching



Why Demonstratives?

- ❖ Demonstratives are available in both Mandarin and English, but they function differently in the DP domain: Mandarin demonstrative has a unique behavior, functioning similarly to English definite determiners (e.g., Saha et al 2024; Dayal & Jiang 2021; Jenks 2018, etc).
 - In Mandarin, the demonstrative functions as the preferred anaphoric determiner
 - In English, the demonstrative determiner is not preferred in anaphoric contexts
- ❖ By focusing on the demonstrative, we examine an interesting asymmetry resulting from the different grammatical constraints on the use of demonstrative determiners in the two languages.



Methodological Considerations for Acceptability Judgments of Code-switching (González-Vilbazo et al, 2013)

- ❖ Lexical Item Vary in Reasons to Code-switch
 - Our solution: carefully chosen lexical items from corpus studies, sources from the internet (blog), bilingual speaker suggestions: e.g. English *quiz*, Mandarin 神仙 'Chinese god'
- ❖ Modality of Presentation: Code-switching often auditory
 - Our solution: Although we kept written stimuli for consistency with experimental semantics work, we began with several training trials to build confidence in the rating of code-switched written stimuli
- ❖ Overall stigma of code-switching, continuum of language modes
 - Our Solution: training sessions with code-switched stimuli (half of our experiments were Mandarin based, the other half English-based); instructions themselves included parts that were in Mandarin and parts that were in English to signal that this is a bilingual environment



Experimental Design

- ❖ Participants (N=60) were recruited from Prolific. Co
- ❖ Half (N=30) of the participants are recruited for the Mandarin-to-English code-switching survey
 - Screeners: Native Language: **Mandarin**, Fluent Language: **English**, Self-identified Bilinguals
 - Instructions were a balanced (50%/50%) mix of Mandarin and English
 - Target items all involved **Mandarin as matrix language**, NPs code-switched to English
- ❖ Half (N=30) of the participants are recruited for the English-to-Mandarin code-switching survey
 - Screeners: Native Language: **English**, Fluent Language: **Mandarin**, Self-identified Bilinguals
 - Instructions were a balanced (50%/50%) mix of Mandarin and English (exact same as above)
 - Target items all involved **English as matrix language**, NPs code-switched to Mandarin



Mandarin Matrix Survey: 那个 quiz vs. quiz

- ❖ Following prior work on preferences in monolingual Mandarin and English (Saha et al., 2023, 2024), participants view a context sentence and then rate paired continuations with Mandarin demonstrative noun phrases and bare noun phrases.
- ❖ The code-switched DPs are at object position in all context sentences and subject position in all continuation sentences
- ❖ Most of the DPs are extracted from the Mandarin-English Code-Switching in South-East Asia corpus to ensure the naturalness of the lexical items

Screenshot from Mandarin-to-English survey

Q1_Man

</> *

李四有一个assignment，下周要交给老师。

最不自然

最自然

那个assignment很
难。

assignment很难。



English Matrix Survey: the 神仙 vs that 神仙

- ❖ Context-continuation pairs with English definite and demonstrative noun phrases
- ❖ The code-switched DPs are at object position in all context sentences and subject position in all continuation sentences
- ❖ Most of the DPs are extracted from online blogs to ensure the naturalness of the lexical items

Screenshot from English-to-Mandarin survey

Q1_ENG

</> ☆

I need to write a 议论文。

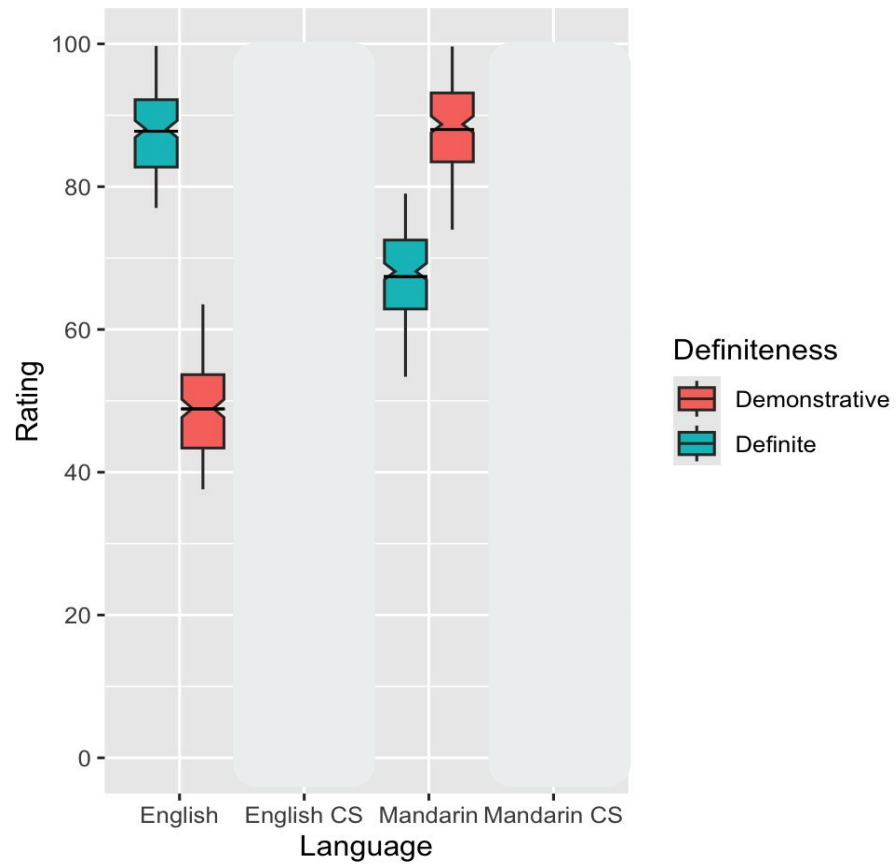
Least Natural

Most Natural

The 议论文 has to
be more than 2000
words.

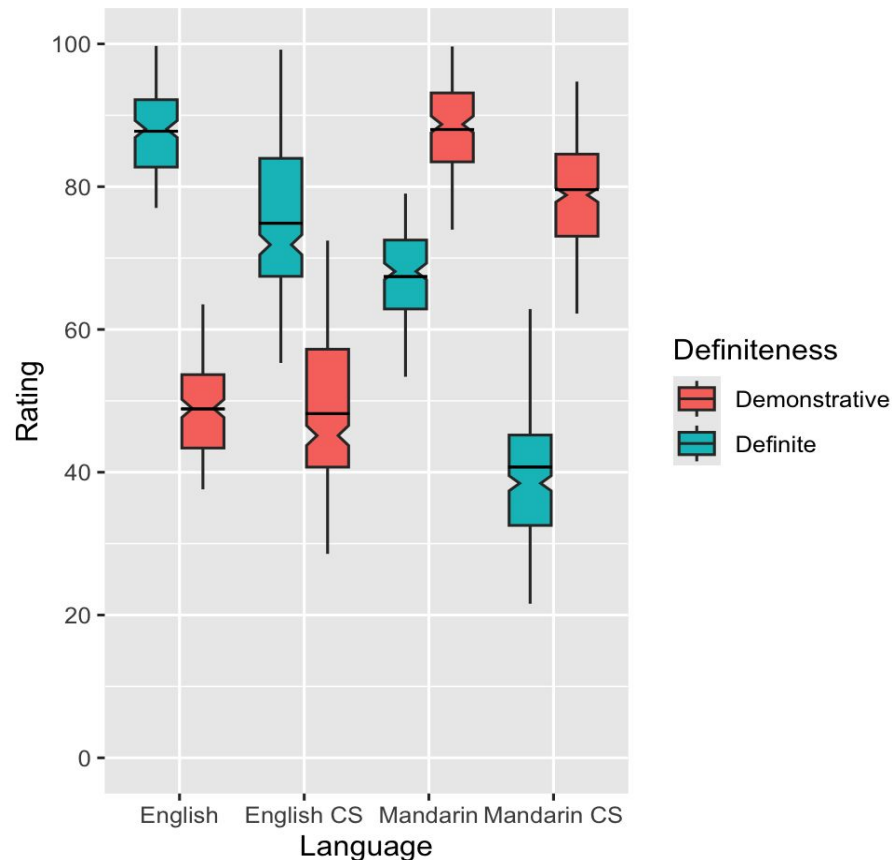
That 议论文 has to
be more than 2000
words.

Results



Results

- ❖ Strong effect of matrix language on acceptability, interaction of matrix language and definiteness
- ❖ In fact, the patterns in the code-switched conditions are very similar to the corresponding monolingual conditions
 - English definite > English demonstrative
 - Mandarin demonstrative > Mandarin bare noun





Interim Conclusions, follow-ups

- ❖ The result aligns with the predictions of the Matrix Language Framework when it comes to code-switching within the DP domain because
 - D head provided by the matrix language (in Mandarin: *nage*; in English: *the*) is preferred
 - Changing the direction of the code-switching has an effect on the speakers' preferences
- ❖ Other questions arise since the full paradigm contains more items (*dem/def* in each language):
 - *nage* quiz
 - *ø* quiz
 - *that* quiz
 - *the* quiz

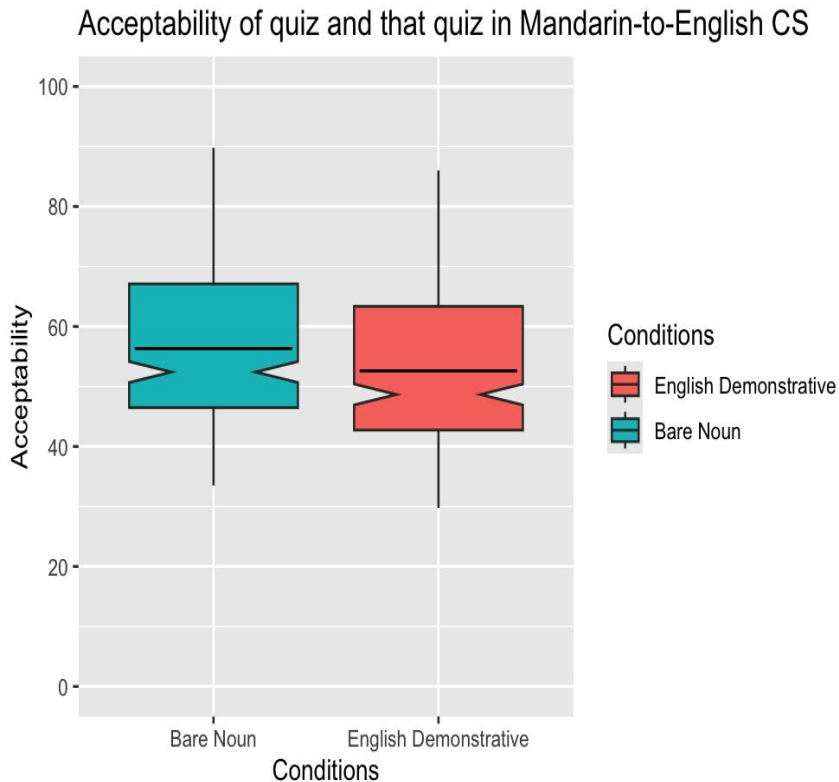


Follow-ups

	Survey 1	Survey 2
Conditions in Mandarin-to-English CS	Bare Noun vs. English Demonstrative	Bare Noun vs. English Determiner
Example	∅ quiz vs. that quiz	∅ quiz vs. the quiz

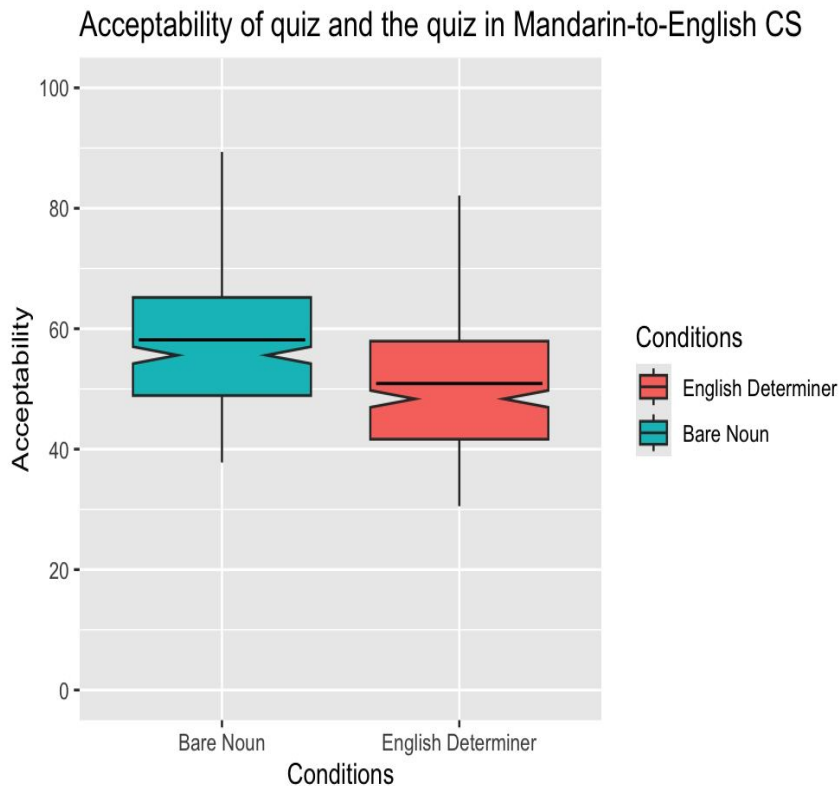
Results: Survey 1

- ❖ No significant effect of Definiteness
 - No significant difference between the acceptability of bare noun and English demonstrative noun phrases in Mandarin-to-English CS



Results: Survey 2

- ❖ The acceptability of bare noun phrases is significantly higher than that of English definite noun phrases ($\beta=-7.256$, $SE=2.240$, $t(643)=-3.24$, $p<0.01$).





Conclusion

- ❖ Acceptability in Mandarin-to-English CS continues to support matrix language affecting the determiner
 - Mandarin demonstrative > Bare Noun
 - Bare Noun ≈ English demonstrative
 - Bare Noun > English definite
- ❖ Additionally, we propose that there are two relevant dimensions of CS: the *code* and the *grammar*
- ❖ Code = the language the item is in (e.g., nage vs. that - both demonstratives but in different languages); Grammar (syn/sem features) = definite vs. demonstrative determiner
 - No CS > No CS (less preferred anaphoric definite)
 - No CS (less preferred anaphoric definite) ≈ CS-ed code
 - No CS (less preferred anaphoric definite) > CS-ed code and grammar
- Code-switching both the code and the grammar could be much more costly than code-switching solely the code or the grammar, leading to lower acceptability



Reference

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- ❖ MacSwan, J. (2005). Codeswitching and generative grammar: A critique of the MLF model and some remarks on “modified minimalism”. *Bilingualism: language and cognition*, 8(1), 1-22.
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